

Team and idea

Team TUF · Track 01 · Automate Workflow with Meetily.ai · SCTCE

Meetily today is audio-only. We add two layers: visual context so it can see the meeting, and meeting assist so it can help in the moment.

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LAYER 1

Visual context - screen/video grounding fused with Meetily's transcript (keyframes, OCR, timeline fusion).

LAYER 2

Meeting assist - Visual Copilot that detects explanatory intent and offers 1-click share of the right diagram.

TRACK FIT

Two workflow layers on Meetily - not a standalone vision app.

Problem statement

Meetily has strong audio context. Meetings also have a screen - and that visual context is missing today.

Meetily today: audio → transcript → summary. Missing: what's on screen / shared video when people say "as seen here."

AUDIO-ONLY LIMIT

What Meetily already does well

Local recording, transcription, and summaries from speech.

What it cannot resolve

Deictic speech tied to slides, dashboards, code, or diagrams - because it never sees them.

WHY VISUALS MATTER LIVE

Notes stay vague

Without screen context, there is only a little context for summarisation.

Need appears mid-sentence

Speaker stops to search/share a diagram - conversation breaks.

Proposed solution

Two layers on Meetily: first give it eyes (visual context), then give it hands (meeting assist).

LAYER 1 · VISUAL CONTEXT

See the meeting
Screen/video keyframes + dHash + local OCR, fused into Meetily’s timestamped transcript.

Grounded notes
Resolves “as seen here” into exact facts (e.g. HTTP 504 12.4% on auth-worker). Passes the clipboard test.

Local synthesis
Ollama cross-references speech against on-screen content for actionable meeting notes.

LAYER 2 · MEETING ASSIST

Detect need
Live Whisper stream + intent cues (“look at this”, explanatory pauses) when a visual would help.

Retrieve visuals
Indexed local repo / vector cache returns 2-3 presentation-grade candidates in a side panel.

1-click assist
Preview, copy, or share to screen share - no tab-switching mid-sentence.

30-hour MVP

Prove the core loop - not every feature.

PHASE 1

Meetily → **app** - pull live/post transcript & context (Agent API / export hooks).

PHASE 2

Context extract - topic, entities, explanatory intent → visual type (diagram / chart / image).

PHASE 3

Local retrieval - small curated /assets set (architecture, algorithms, patterns) via semantic match.

PHASE 4

Copilot UI - suggestion card with Preview + Share; optional screen keyframe + OCR if time allows.

Technology stack

Open-source, local-first components across both feature directions.

MEETILY

Recording, timestamped transcript, summaries, Agent API / MCP hooks

SPEECH

Whisper - live transcription stream feeding the topic extractor

KEYFRAMES

Background observer + difference hashing (dHash) to detect slide/window transitions

OCR

Tesseract OCR on changed frames only

FUSION + LLM

Chronological timeline fusion of OCR + transcript, grounded synthesis via Ollama

RETRIEVAL

Indexed local asset repository / vector cache for diagrams and charts

End-to-end user journey

From live discussion and screen sharing to real-time visual assistance and complete meeting notes.

01

SPEAK

Explains a complex topic or incident.
Meetily transcribes and tracks conversation context live.

02

SHOW

Shares screen with dashboard or slides.
Vision layer captures and reads on-screen content automatically.

03

SUGGEST

Continues talking without searching.
Copilot proactively suggests the matching diagram in side panel.

04

SHARE

Clicks **Share** in one click.
Visual appears live on call — zero awkward pauses or tab switching.

05

SUMMARY

Receives final meeting notes.
Spoken words and screen facts are merged into clear action items.

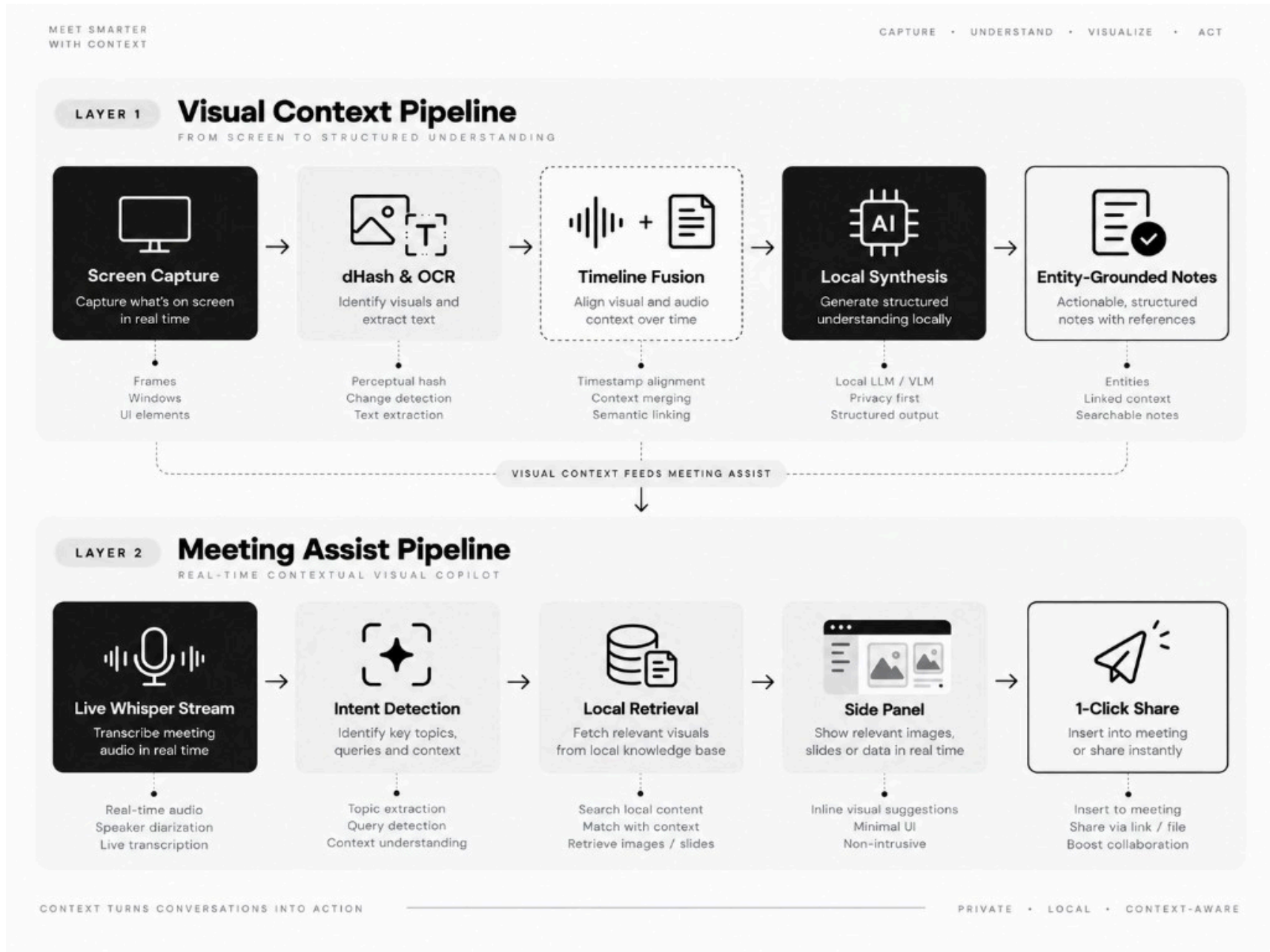
LIVE MEETING ASSIST · DURING CALL

Anticipates when a visual is needed and surfaces the right diagram — 1-click share keeps conversations completely uninterrupted.

SCREEN-GROUNDED NOTES · AFTER CALL

Connects spoken discussion with what was shown on screen — turning vague phrases like “as seen here” into clear, self-contained facts.

Solution flowchart / architecture



Future scaling / scope

From Visual Copilot → full meeting action agent.

TODAY

Live visual suggestions from Meetily context (Preview + Share).

NEXT

Meeting-grounded slide packs and automatic visual documentation.

THEN

Cross-tool automation: Meetily → visuals → slides → docs → Slack/Discord → follow-up.

VISION

Meeting intelligence that doesn't only remember - it turns conversation into action.